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SUPERSEDING
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February 4, 1977

MILITARY SPECIFICATION

CONDUIT, AND CONDUIT FITTINGS, METAL, RIGID, POLYVINYL CHLORIDE PVC-COATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrical conduit and conduit fittings, metal, rigid, polyvinyl chloride (PVC) coated, for use in corrosive environments.

1.2 Classification.

1.2.1 Types. When no type is specified, standard length conduit with one coupling attached shall be furnished. Conduit fittings shall be the types as specified in applicable Federal and Military specifications and standards (see 6.2).

1.2.2 Sizes. The conduit and conduit fittings shall have the following nominal or trade thread sizes in inches as specified (see 6.2):

1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 4-1/2, 5, and 6.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5975

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SPECIFICATIONS

FEDERAL

W-C-586 - Conduit Outlet Boxes, Bodies, and Entrance Caps, Electrical: Cast Metal - For Shore Use.
W-F-408 - Fitting for Conduit, Metal, Rigid, (Thick-wall and Thin-wall (EMT) Type).
WW-C-581 - Conduit, Metal, Rigid and Intermediate; and Coupling, Elbow, and Nipple, Electrical Conduit, Steel, Zinc-Coated.

MILITARY

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of.
MIL-F-28675 - Fittings, Conduit, Explosion Proof (Couplings, Seals, Unions, Elbows).

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents that are indicated as DoD-adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

C80.1 - Specification for Rigid Steel Conduit, Zinc-Coated.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B117 - Salt Spray (Fog) Testing.
D149 - Dielectric Breakdown Voltage and Dielectric Strength of Electric Insulating Materials at Commercial Power Frequencies.
D638 - Tensile Properties of Plastics.
D1790 - Brittleness Temperature of Plastic Film by Impact.

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D2240 - Indentation Hardness of Rubber and Plastics by Means of a Durometer.
G6 - Abrasion Resistance of Pipeline Coatings.
G11 - Effects of Outdoor Weathering on Pipeline Coatings.
G12 - Nondestructuve Measurement of Film Thickness of Pipeline Coatings on Steel.
G21 - Resistance of Synthetic Polymeric Materials to Fungi, Recommended Practice for Determining.
G23 - Operating Light- and Water-Exposure Apparatus (Carbon-Arc Type) for Exposure of Nonmetallic Materials.
G29 - Algae Resistance of Plastic Films.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

RN1 - Polyvinyl-Chloride Externally Coated Galvanized Rigid Steel Conduit and Electrical Metallic Tubing.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL6 - Rigid Metal Conduit.
UL514 - Outlet Boxes and Fittings.
UL866 - Outlet Boxes and Fittings, Electrical, for Use in Hazardous Locations, Class I, Groups A, B, C, and D and Class II, Groups E, F, and G.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. This specification covers PVC externally coated, galvanized, rigid steel conduit and fittings. Unless otherwise specified herein, the PVC-coated conduit and conduit fittings shall conform to NEMA RN1 for type 40, PVC coatings.

3.2 First article. When specified (see 6.2), the contractor shall furnish one conduit and one fitting for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Standard commercial product. The conduit and fittings shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better

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features that are not specifically prohibited by this specification, but that are a part of the manufacturer's standard commercial product, shall be included in the conduit and fittings being furnished. A standard commercial product is a product that has been sold or is being currently offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures and represents the latest production model(s).

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Materials. Materials used shall be free from defects that would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials that have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect that reduces the ability of the castings to perform its intended function.

3.7 Metal conduit and fittings.

3.7.1 Conduit. The galvanized steel conduit, upon which the PVC coating is applied, shall conform to applicable sections of WW-C-581, ANSI C80.1, and UL6. The conduit shall be hot-dipped galvanized on all surfaces except that equivalent protection to zinc coating may be provided for internal surfaces. A PVC-coated coupling shall be furnished with each length of conduit.

3.7.2 Conduit fittings. Conduit fittings (see 6.4.1) on which the PVC coating is applied shall conform to applicable Federal and Military specifications and standards (see 1.2.1 and 6.2.1), including as applicable W-F-408, W-C-586, WW-C-581, and MIL-F-28675. Conduit fittings shall also conform to applicable UL standards, including UL514 and UL886.

3.8 PVC coating application. The coating applied to the metal conduit and fittings shall be domestic, virgin, homo-polymer PVC conforming to NEMA RN1.

3.8.1 Surface preparation. The interior and exterior of all surfaces to be coated shall be cleaned of grease, oil, dirt, mill scale, oxide, or other foreign matter in such a manner that the galvanized surface is not harmed or eroded. The galvanized metal surfaces shall be etched or roughened to increase the contact area and provide better adhesion for the primer coat.

3.8.2 Surface primer. The cleaned metal interior and exterior surfaces shall be primed with a heat-polymerizing, adhesive primer suitable for use with the PVC coating material to be applied. The primer shall be bonded to

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the galvanized metal surface by heating the metal 300° to 475° Fahrenheit (F) in an oven. The bonded primer coating shall be no less than 1/2-mil (0.0005-inch) thick. Surfaces shall be primed and coated as soon as possible after being cleaned to prevent the formation of rust or oxide on the cleaned metal.

3.8.3 Heat treatment. During or after the metal-priming process, the metal shall be preheated just prior to receiving the PVC coating. After being coated, the metal shall be post-cured in an oven that bonds the coating to the metal surface and removes surface flaws such as pinholes and thin spots.

3.8.4 Coating application. The coating shall be applied by spraying, electrostatic or cloud-chamber deposition, fluidized-bed technique, or plastisol immersion.

3.8.5 Threads. All threaded portions of conduit and fittings shall be uncoated or otherwise cleaned of the PVC coating after the coating process. All threads shall have thread protectors installed after completion of the coating operation.

3.8.6 Inside surfaces. The inside surfaces of the PVC-coated conduit and those conduit fittings used as a portion of a wire or cable raceway shall not be coated with any PVC material.

3.9 Coating physical characteristics. Coating characteristics described herein apply to the PVC coating as deposited on the metal substrate unless otherwise indicated.

3.9.1 Uniformity and thickness. The PVC coating shall have a nominal thickness of 0.040 inch with a minimum thickness of 0.035 inch on conduit and a minimum thickness of 0.040 inch on fittings at any point of the coated metal surface when tested in accordance with 4.5.1. The coating shall be continuous over the entire unthreaded portion of the metal surface and shall be chambered at the junction of the threaded and unthreaded surfaces.

3.9.2 Adhesion. The adhesion of the PVC coating to the metal surface shall be greater than the cohesive strength of the coating when tested in accordance with 4.5.2.

3.9.3 Hardness. The hardness of the PVC coating shall be greater than 75 as measured on the Shore A Durometer when tested in accordance with 4.5.3.

3.9.4 Tensile strength. The tensile strength of the PVC coating shall be greater than 2,000 pounds per square inch when tested in accordance with 4.5.4.

3.9.5 Elongation. The elongation of the PVC coating shall be not less than 200 percent when tested in accordance with 4.5.5.

3.9.6 Abrasion resistance. The abrasion resistance shall be greater than 200 hours with no failure when tested in accordance with 4.5.6.

3.9.7 Bendability. The bent specimen shall show no deleterious effects on the coating such as wrinkles, blisters, peeling, lessening of adhesion, and holidays (see 3.10.2) when tested in accordance with 4.5.7.

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3.9.8 Environmental. The PVC coating shall undergo no deleterious effects or exhibit signs of deterioration including holidays, undercutting, discoloration, blisters, chalkiness, peeling, flaking, cracking, tackiness, loss of gloss, and brittleness. When specified (see 6.2), the PVC coating shall be tested in accordance with 4.5.8.

3.9.9 Fungus resistance. The PVC coating shall be resistant to fungus growth. When tested in accordance with 4.5.9, there shall be no traces of fungus growth.

3.9.10 Algae resistance. The PVC coating shall be resistant to algae growth. When tested in accordance with 4.5.9, there shall be no traces of algae growth.

3.9.11 Temperature.

3.9.11.1 Operating temperature. The PVC coating shall be capable of continuous service without significant loss in the coating life expectancy at temperatures from -5°F to +160°F.

3.9.11.2 Brittleness temperature. The free-film brittleness temperature of the PVC coating shall be less than 5°F when tested in accordance with 4.5.10.

3.10 Coating electrical characteristics.

3.10.1 Dielectric strength. The dielectric strength of the PVC coating shall be greater than 400 volts (V) per mil thickness at 60 Hertz in accordance with 4.5.11.

3.10.2 Holidays. The PVC coating shall be free of holidays (see 6.4.2) when tested in accordance with 4.5.12. An electrical discharge through the coating, detected visually, aurally, or by instrument, shall be considered evidence of an unacceptable flaw in the coating.

3.11 Conduit fittings.

3.11.1 Hardware. All nuts, bolts, screws, washers, and other small hardware used to assemble PVC-coated fittings shall be stainless steel and shall have screw threads in accordance with FED-STD-H28. Screw and bolt heads and nuts shall have captive washers or otherwise shall be so designed such that when the screws, bolts, or nuts are tightened, the PVC coating will not be cut, torn, or otherwise injured such that the seal between screw or bolt and fitting is impaired. Truss head screws may be used. All stainless steel fasteners shall be encapsulated with a corrosion-resistant plastic coating.

3.11.2 Extension sleeves. All female threaded portions of fittings shall have the PVC coating extended beyond the end of the threaded portion in such a manner to form a flexible PVC sleeve that will overlap the conduit or mating male fitting designed to be inserted into the female threads. The length of the sleeve extension shall be not less than the outside diameter (OD) of the conduit size for which the fitting is designed for conduit sizes from 1/2 to 2 inches. For conduit sizes larger than 2 inches, the sleeve extension shall be not less than 2 inches. The PVC sleeve shall have a nominal thickness of 0.040 inch with a minimum thickness of 0.035 inch. The inside diameter of the

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sleeve shall be less than the OD of the PVC-coated conduit or mating male fitting to insure a tight fit of the sleeve onto the mating conduit or fitting. The sleeve shall be resilient enough to stretch over the coating on the conduit and form a vapor-tight joint without the use of patching compound. All sleeves shall be square cut.

3.12 PVC coating patching compound. When specified (see 6.2), a brush-type liquid compound that will bond, patch, and seal the process-applied PVC coating shall be provided. Spray-type patching material shall not be permitted. The size of the patching compound container shall be as specified (see 6.2). Two different compounds may be furnished, in which case one compound shall be used as a sealer and the other compound shall be used as a patching substance. The patching compound shall be compatible with the PVC coating and shall be manufactured by the same manufacturer as the PVC-coated conduit. When specified (see 6.2), the patching compound shall be tested in accordance with 4.5.12.

3.12.1 Properties. The patching compound shall be used as a joint sealer and as a flaw-patching substance. It shall be capable of permanently bonding the PVC sleeves of some types of conduit fittings to the outer PVC jacket and shall be capable of repairing damaged PVC-coated areas. Adhesive properties shall be as specified in 3.9.2.

3.12.2 Characteristics. The patching compound shall have physical, chemical, and electrical characteristics similar to or better than those characteristics of the process-applied PVC coating specified herein when applied according to the manufacturer's instructions.

3.12.3 Application instructions. Instructions for applying the patching compound shall be printed or otherwise legibly marked upon the patching compound container. The instructions on the container shall include storage, handling, safety, and area coverage of contents.

3.13 Fire and casualty hazards.

3.13.1 Each contractor shall submit, to the contracting agency, proof that the conduit and conduit fittings proposed for supply by the contractor under this specification conform to the requirements of applicable UL standards. The listing mark of UL may be accepted as evidence that the conduit and conduit fittings conform to this requirement.

3.13.2 In lieu of the UL listing mark, the bidder may submit independent proof, satisfactory to the contracting agency, that the bidders' conduit and conduit fittings conform to the applicable requirements of the published UL standards including methods of tests.

3.13.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of the bidders' materials.

3.14 Workmanship. The PVC coating shall be smooth and free of flaws, including cuts, tears, blisters, bubbles, pinholes, thin spots, and other surface defects. All threaded portions of conduit and conduit fittings shall be clean and undamaged.

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3.15 Marking. The PVC coating on conduits shall be either permanently marked or permanently labeled with the manufacturer's name and type number in a clear and legible manner that will not injure the PVC coating.

3.16 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on each type of conduit and conduit fitting when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4, the tests of 4.5, and, when specified, the first article packaging inspection (see 4.6 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.4 and the tests of 4.5.1, 4.5.2, 4.5.12, and the packaging inspection of 4.6. When specified (see 6.2), all the tests of 4.5, except the environment test of 4.5.8 and the patching compound test of 4.5.13, shall be performed.

4.3 Sampling. Sampling shall be in accordance with MIL-STD-105. All conduit and conduit fittings offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level II and the acceptable quality level (AQL) shall be 4.0 percent defective. If an inspection lot is rejected, the contractor may rework it to correct the defects

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or screen out the defective units and resubmit for a complete reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.4 Examination. Each item selected shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests. Failure of the conduit and conduit fittings to meet the requirements of the tests in 4.5 shall constitute cause for rejection.

4.5.1 Uniformity and thickness. The PVC coating uniformity and thickness shall be tested in accordance with ASTM G12 for conformance to 3.9.1.

4.5.2 Adhesion tests. The adhesion of the PVC coating shall be determined by making two parallel cuts 0.5 inch apart and not less than 0.75 inch long at a suitable location on a suitable size PVC-coated conduit and conduit fitting. A third cut shall be made at right angles to the first cuts and crossing both. The cuts shall be made with a sharp knife and shall penetrate through the PVC film to the substrate. The knife shall then be worked under the coating for 0.5 inch to free the plastic from the metal. The plastic tab thus formed shall be pulled perpendicularly to the metal surface. The plastic shall then tear apart leaving particles of plastic clinging to the metal surface, indicating that the adhesion is greater than the tensile strength of the PVC coating for conformance to 3.9.2. When a first article test is specified (see 4.2.1 and 6.3), the adhesion test shall be performed on the test specimen after the environmental test (see 4.5.8) has been completed.

4.5.3 Hardness test. The PVC coating hardeners shall be tested in accordance with ASTM D2240 for conformance with 3.9.3.

4.5.4 Tensile strength test. The tensile strength test of PVC coating shall be in accordance with ASTM D638 for conformance with 3.9.4.

4.5.5 Elongation test. The elongation test of PVC coating shall be in accordance with ASTM D638 for conformance with 3.9.5.

4.5.6 Abrasion resistance test. The PVC coating shall be tested for abrasion resistance in accordance with ASTM G6 for conformance with 3.9.6.

4.5.7 Conduit bendability test. At a temperature of 73°F $\pm$2°F, the PVC-coated conduit shall be capable of being bent into a quarter of a circle by a commercial rigid-conduit bender designed for use on the next larger size of rigid conduit for conformance to 3.9.7. For example, 1/2-inch PVC-coated conduit shall be capable of being bent by a 3/4-inch rigid-conduit bender. After the bendability test, the adhesion test (see 4.5.2) and holiday test (see 4.5.12) shall be performed on the test specimen. When a first article is specified (see 4.2.1 and 6.3), the bendability test shall be performed on the test specimen after the environmental test (see 4.5.8) has been completed.

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4.5.8 Environmental test(s). The environmental test(s) shall consist of either the marine environment test (see 4.5.8.1) or the salt spray test (see 4.5.8.2) and the artificial weathering test (see 4.5.8.3). The patching compound test (see 4.5.13) shall be performed in conjunction with the environmental test for conformance with 3.9.8. Not less than two test specimens for each type of item shall be tested. One of the test specimens of each type shall be scribed, to the depth of the underlying metal substrate, in not less than two separate areas. The width of each scribed exposed metal area on the test specimen shall be not less than 1/8 inch. The length of each scribed area shall be not less than one-third the largest dimension except that conduit and nipples shall be scribed circumferentially not less than one-third the circumference in arc length.

4.5.8.1 Marine environment test. The PVC-coated conduit and fittings shall be tested in accordance with ASTM G11, as applicable, in a tropical corrosive marine environment for a period of 12 months. Conduit and fittings shall be mounted on standard ASTM 45° angle racks. The racks shall face the prevailing wind and shall be located within 100 feet from the ocean surf at high tide, such that the prevailing wind deposits salt spray continuously on the test specimens. During and after testing, the PVC-coated specimens shall be examined for compliance with 3.9.8.

4.5.8.2 Salt spray test. The PVC-coated conduit and conduit fittings shall be tested in accordance with ASTM B117 for not less than 3,000 hours. After testing, the test specimens shall be examined for compliance with 3.9.8.

4.5.8.3 Artificial weathering. The PVC-coated conduit and conduit fittings shall be tested in accordance with ASTM G23 for not less than 4,000 hours. After testing, the test specimens shall be examined for compliance with 3.9.8.

4.5.9 Fungus and algae resistance. The PVC coating shall be tested for fungus and algae resistance in accordance with ASTM G21 and ASTM G29 respectively for conformance to 3.9.9 and 3.9.10 respectively.

4.5.10 Brittleness temperature test. The brittleness temperature test of PVC coating shall be in accordance with ASTM D1790 for conformance with 3.9.11.2.

4.5.11 Dielectric strength test. The dielectric strength test of PVC coating shall be in accordance with ASTM D149 for conformance with 3.10.1.

4.5.12 Holiday tests. The entire surface of the PVC coating shall be tested for holidays (see 6.4.2) by using a high-potential holiday detector that impresses 8000 V across the coating for conformance with 3.10.2. One electrode of the tester shall maintain complete circumferential contact with the coating while transversing the entire length of the coating system, and the other electrode shall be the underlying metal substrate.

4.5.13 Patching compound test. This test shall be performed on the exposed metal surface of the PVC coating formed by making a rectangular pene- trating cut 0.5-inch wide by 0.75-inch long in the PVC coating and removing the cut PVC coating material. The patching compound shall be applied as minimum as 0.040-inch thick to the metal surface in accordance with the instructions on the container. The patching compound shall overlap the rectangular exposed

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metal surface by at least 0.25 inch on all sides. The environmental test (see 4.5.8) shall be performed on the patched area after the patching compound has been allowed to cure for conformance with 3.12.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-E-17555. The inspection shall consist of the quality conformance inspection and, when specified (see 6.2), a first article shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-E-17555 with the levels of preservation and packing as specified (see 6.2).

6. NOTES

6.1 Intended use. PVC-coated, galvanized, rigid steel conduit and conduit fittings are used as electrical wire or cable raceways in corrosive environments.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of conduit fittings and applicable specification or standard defining type (see 1.2.1).
c. Size of conduit and conduit fittings (see 1.2.2).
d. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
e. When marine environment or salt spray and the artificial weathering tests are required (see 3.9.8).
f. When patching compound is required and size of patching compound container (see 3.12).
g. When the patching compound test is required (see 3.12).
h. When all the tests of 4.5, except for 4.5.8 and 4.5.13, shall be performed (see 4.2.2).
i. When first article packaging inspection is required for examination and test and the time frame required for submission (see 4.6).
j. Levels of preservation and packing required (see 5.1).

6.2.1 Contract data requirements. When this specification is used in a procurement that incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete conduit or conduit fitting unit or it may be a standard production

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item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all acquisition instruments regarding arrangement for examinations, tests, and approval of the first article.

6.4 Definitions.

6.4.1 Rigid conduit fittings. Rigid conduit fittings are defined in this specification to include, but are not limited to, the following fittings: conduit couplings, elbows, unions, reducers, plugs, adapters, service entrance heads, seals, clamps, breathers and drains, expansion joints, conduit hangers, conduit outlet bodies, cast device boxes, junction boxes, cable and cord fittings, connectors, cover plates, and lighting fixtures and hangers.

6.4.2 Holiday. A holiday is defined as a surface coating flaw that permits current drainage through the PVC coating to the metal substrate.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - ER
Navy - YD
Air Force - 85

Preparing Activity:

Navy - YD

Project No. 5975-0538

Review activities:

Air Force - 99
DLA - GS

User activities:

Army - ME
Navy - MC

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